

Subject : Physics,
Chemistry,
Biology

NEET Grand Test 1

Standard : 11,12

Total Mark : 720

Paper Set : 1

Date : 15-06-2026

Time : 3H:15M

Physics - Section A (MCQ)

- (1) A solid sphere of radius R is placed on a smooth horizontal surface. A horizontal force F is applied at a height h from the lowest point. For the maximum acceleration of the centre of mass, which is correct?

- (1) $h = R$
(2) no relation between h and R
(3) $h = 0$
(4) $h = 2R$

- (2) During the melting of a slab of ice at 273 K at atmospheric pressure,

- (1) Internal energy of ice-water system remains unchanged.
(2) Positive work is done by the ice-water system on the atmosphere.
(3) Internal energy of the ice-water system decreases.
(4) Positive work is done on the ice-water system by the atmosphere.

- (3) n moles of an ideal gas with constant volume heat capacity C_v undergo an isobaric expansion by a certain volume. The ratio of the work done in the process to the heat supplied is

- (1) $\frac{nR}{C_v + nR}$ (2) $\frac{nR}{C_v - nR}$
(3) $\frac{nR}{C_p}$ (4) $\frac{nR}{C_v}$

- (4) The least count of a screw gauge is 0.01 mm . If the pitch is increased by 75% and the number of divisions on the circular scale is reduced by 50%, the new least count will be $\dots \times 10^{-3}\text{ mm}$.

- (1) 25 (2) 35
(3) 15 (4) 55

- (5) The velocity of a small ball of mass 0.3 g and density 8 g/cc when dropped in a container filled with glycerine becomes constant after some time. If the density of glycerine is 1.3 g/cc , then the value of the viscous force acting on the ball will be $x \times 10^{-4}\text{ N}$. The value of x is [use

$g = 10\text{ m/s}^2$]. (in .125)

- (1) 24 (2) 23

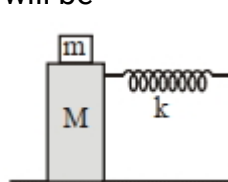
- (3) 25 (4) 22

- (6) A ball is dropped from a height of 20 m . If the coefficient of restitution for the collision between ball and floor is 0.5 , after hitting the floor, the ball rebounds to a height of $\dots\text{ m}$.

- (1) 4 (2) 3

- (3) 5 (4) 2

- (7) In the given figure, a mass M is attached to a horizontal spring which is fixed on one side to a rigid support. The spring constant of the spring is k . The mass oscillates on a frictionless surface with time period T and amplitude A . When the mass is at its maximum displacement (extreme position), another mass m is gently placed upon it. The new amplitude of oscillation will be



- (1) $A\sqrt{\frac{M-m}{M}}$ (2) $A\sqrt{\frac{M}{M+m}}$

- (3) $A\sqrt{\frac{M+m}{M}}$ (4) $A\sqrt{\frac{M}{M-m}}$

- (8) When milk is churned, cream gets separated due to

- (1) gravitational force (2) frictional force
(3) centrifugal force (4) centripetal force

- (9) Two plates A and B have thermal conductivities $84\text{ W m}^{-1}\text{ K}^{-1}$ and $126\text{ W m}^{-1}\text{ K}^{-1}$ respectively. They have the same surface area and same thickness. They are placed in contact along their surfaces. If the temperatures of the outer surfaces of A and B are kept at 100°C and 0°C respectively, then the temperature of the surface of contact in steady state is $\dots^\circ\text{C}$.

- (1) 20 (2) 40

- (3) 60 (4) 80

- (10) If a satellite orbiting the Earth is 9 times closer to the Earth than the Moon, what is the time

period of rotation of the satellite? Given rotational time period of Moon = 27 days and gravitational attraction between the satellite and the moon is neglected.

- (1) 1 day (2) 81 days
(3) 27 days (4) 3 days

- (11) The Young's modulus of steel is twice that of brass. Two wires of the same length and of the same area of cross-section, one of steel and another of brass, are suspended from the same roof. If we want the lower ends of the wires to be at the same level, then the weights added to the steel and brass wires must be in the ratio of

- (1) 2 : 1 (2) 1 : 2
(3) 1 : 1 (4) 4 : 1

- (12) An object is projected in the air with initial velocity u at an angle θ . The projectile motion is such that the horizontal range R is maximum. Another object is projected in the air with a horizontal range half of the range of the first object. The initial velocity remains the same in both cases. The value of the angle of projection, at which the second object is projected, will be degrees.

- (1) 85 (2) 80
(3) 15 or 75 (4) 70

- (13) There are two vessels filled with an ideal gas where the volume of one is double the volume of the other. The large vessel contains the gas at 8 kPa at 1000 K , while the smaller vessel contains the gas at 7 kPa at 500 K . If the vessels are connected to each other by a thin tube allowing the gas to flow and the temperature of both vessels is maintained at 600 K , at steady state the pressure in the vessels will be (in kPa).

- (1) 4.4 (2) 6
(3) 24 (4) 18

- (14) The fundamental frequency of a closed pipe is equal to the frequency of the second harmonic of an open pipe. The ratio of their lengths is

- (1) 1 : 2 (2) 1 : 4
(3) 1 : 8 (4) 1 : 16

- (15) What will be the ratio of the distance moved by a freely falling body from rest in 4^{th} and 5^{th} seconds of journey?

- (1) 7 : 9 (2) 4 : 5
(3) 16 : 25 (4) 1 : 1

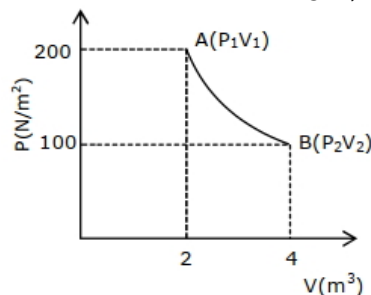
- (16) A 100 g of iron nail is hit by a 1.5 kg hammer striking at a velocity of 60 ms^{-1} . What will be the rise in the temperature of the nail in $^{\circ}\text{C}$ if one-fourth of the energy of the hammer goes into heating the nail? [Specific heat capacity of

iron = $0.42 \text{ Jg}^{-1}\text{C}^{-1}$]

- (1) 675 (2) 1600
(3) 16.07 (4) 6.75

- (17) One mole of an ideal gas at 27°C is taken from A to B as shown in the given PV indicator diagram. The work done by the system will be $\times 10^{-1} \text{ J}$.

[Given: $R = 8.3 \text{ J/mole} \cdot \text{K}$, $\ln 2 = 0.6931$] (Round off to the nearest integer)



- (1) 30012 (2) 2147
(3) 834 (4) 17258

- (18) If $|\vec{A} + \vec{B}| = |\vec{A}| + |\vec{B}|$, then the angle between $\vec{A}$ and $\vec{B}$ will be $^{\circ}$.

- (1) 90 (2) 120
(3) 0 (4) 60

- (19) The atomic hydrogen emits a line spectrum consisting of various series. Which series of hydrogen atomic spectra lies in the visible region?

- (1) Brackett series (2) Paschen series
(3) Lyman series (4) Balmer series

- (20) Two statements are given below:

A. When the forward bias voltage across a $p - n$ junction diode increases above a certain threshold voltage, the diode current increases significantly.

B. This current is called reverse saturation current.

Choose the correct answer from the options given below:

- (1) Both Statements A and B are true
(2) Both Statements A and B are false
(3) Statement A is true, but Statement B is false
(4) Statement A is false, but Statement B is true

- (21) The angle of incidence at which reflected light is totally polarized for reflection from air to glass (refractive index n) is

- (1) $\sin^{-1}(n)$ (2) $\sin^{-1}(1/n)$
(3) $\tan^{-1}(1/n)$ (4) $\tan^{-1}(n)$

- (22) Consider a circular coil of wire carrying constant current I , forming a magnetic dipole. The magnetic flux through an infinite plane that contains the circular coil and excluding the

circular coil area is given by ϕ_i . The magnetic flux through the area of the circular coil area is given by ϕ_0 . Which of the following options is correct?

- (1) $\phi_i = -\phi_0$ (2) $\phi_i = \phi_0$
(3) $\phi_i < \phi_0$ (4) $\phi_i > \phi_0$

(23) A galvanometer has a coil of resistance $100\ \Omega$ and gives a full-scale deflection for 30 mA current. If it is to work as a voltmeter of 30 V range, the resistance required to be added will be..... Ω

- (1) 900 (2) 1800
(3) 500 (4) 1000

(24) In a common base amplifier, the phase difference between the input signal voltage and the output voltage is

- (1) 0 (2) $\pi/4$
(3) $\pi/2$ (4) π

(25) Match List-I with List-II.

List-I	List-II
(a) 10 km height over earth's surface	(i) Thermosphere
(b) 70 km height over earth's surface	(ii) Mesosphere
(c) 180 km height over earth's surface	(iii) Stratosphere
(d) 270 km height over earth's surface	(iv) Troposphere

- (1) (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i)
(2) (a) – (i), (b) – (iv), (c) – (iii), (d) – (ii)
(3) (a) – (iii), (b) – (ii), (c) – (i), (d) – (iv)
(4) (a) – (ii), (b) – (i), (c) – (iv), (d) – (iii)

(26) The radii of curvature for a thin convex lens are 10 cm and 15 cm respectively. The focal length of the lens is 12 cm . The refractive index of the lens material is:

- (1) 1.2 (2) 1.4
(3) 1.5 (4) 1.8

(27) A magnetic needle suspended parallel to a magnetic field requires $\sqrt{3}\text{ J}$ of work to turn it through 60° . The torque needed to maintain the needle in this position will be..... J

- (1) $2\sqrt{3}$ (2) 3
(3) $\sqrt{3}$ (4) 1.5

(28) Which of the following statements is false?

- (1) Wheatstone bridge is the most sensitive when all the four resistances are of the same order of magnitude.
(2) In a balanced Wheatstone bridge, if the cell and the galvanometer are exchanged, the null point is disturbed.
(3) A rheostat can be used as a potential divider.
(4) Kirchhoff's second law represents energy conservation.

(29) An electric dipole with dipole moment $5 \times 10^{-6}\text{ Cm}$ is aligned with the direction of a uniform electric field of magnitude $4 \times 10^5\text{ N/C}$. The dipole is then rotated through an angle of 60° with respect to the electric field. The change in the potential energy of the dipole is (in J)

- (1) 0.8 (2) 1.0
(3) 1.2 (4) 1.5

(30) The energy released by the fission of one uranium atom is 200 MeV . The number of fissions per second required to produce 3.2 W of power is (Take $1\text{ eV} = 1.6 \times 10^{-19}\text{ J}$).

- (1) 10^7 (2) 10^{10}
(3) 10^{15} (4) 10^{11}

(31) A parallel-plate capacitor of capacitance $40\ \mu\text{F}$ is connected to a 100 V power supply. Now, the intermediate space between the plates is filled with a dielectric material of dielectric constant $K = 2$. Due to the introduction of the dielectric material, the extra charge and the change in the electrostatic energy in the capacitor, respectively, are:

- (1) 2 mC and 0.2 J (2) 8 mC and 2.0 J
(3) 4 mC and 0.2 J (4) 2 mC and 0.4 J

(32) An alternating current is flowing through a series $L - C - R$ circuit. It is found that the current reaches a value of 1 mA at both 200 Hz and 800 Hz frequency. What is the resonance frequency of the circuit (in Hz)?

- (1) 600 (2) 300
(3) 500 (4) 400

(33) The number of photoelectrons emitted for light of a frequency ν (higher than the threshold frequency ν_0) is proportional to

- (1) threshold frequency ν_0
(2) intensity of light
(3) frequency of light ν
(4) $\nu - \nu_0$

(34) Given below are two statements: one is labelled as Assertion A and the other is labelled as

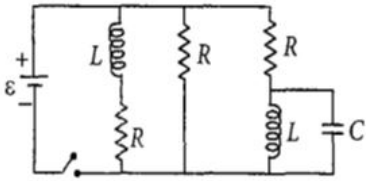
Reason R .

Assertion A : EM waves used for optical communication have longer wavelengths than that of microwaves employed in $RADAR$ technology.

Reason R : Infrared EM waves are more energetic than microwaves (used in $RADAR$). In the light of the given statements, choose the correct answer from the options given below:

- (1) A is false but R is true.
 (2) A is true but R is false.
 (3) Both A and R are true but R is NOT the correct explanation of A .
 (4) Both A and R are true and R is the correct explanation of A .
- (35) A telescope of aperture $3 \times 10^{-2} m$ diameter is focused on a window at $80 m$ distance fitted with a wire mesh of spacing $2 \times 10^{-3} m$. Given: $\lambda = 5.5 \times 10^{-7} m$, which of the following is true for observing the mesh through the telescope?
 (1) Yes, it is possible with the same aperture size.
 (2) Possible also with an aperture half the present diameter.
 (3) No, it is not possible.
 (4) Given data is not sufficient.
- (36) Two metal wires of identical dimensions are connected in series. If σ_1 and σ_2 are the conductivities of the metal wires respectively, the effective conductivity of the combination is
 (1) $\frac{\sigma_1 \sigma_2}{\sigma_1 + \sigma_2}$ (2) $\frac{2\sigma_1 \sigma_2}{\sigma_1 + \sigma_2}$
 (3) $\frac{\sigma_1 + \sigma_2}{2\sigma_1 \sigma_2}$ (4) $\frac{\sigma_1 + \sigma_2}{\sigma_1 \sigma_2}$
- (37) A green light is incident from the water to the air-water interface at the critical angle (θ). Select the correct statement.
 (1) The spectrum of visible light whose frequency is less than that of green light will come out to the air medium.
 (2) The spectrum of visible light whose frequency is more than that of green light will come out to the air medium.
 (3) The entire spectrum of visible light will come out of the water at various angles to the normal.
 (4) The entire spectrum of visible light will come out of the water at an angle of 90° to the normal.
- (38) The vertical component of the earth's magnetic field is $6 \times 10^{-5} T$ at a place where the angle of dip is 37° . The earth's resultant magnetic field at

that place will be (Given $\tan 37^\circ = \frac{3}{4}$)

- (1) $8 \times 10^{-5} T$ (2) $6 \times 10^{-5} T$
 (3) $5 \times 10^{-4} T$ (4) $1 \times 10^{-4} T$
- (39) X different wavelengths may be observed in the spectrum from a hydrogen sample if the atoms are excited to states with principal quantum number $n = 6$. The value of X is
 (1) 11 (2) 15
 (3) 10 (4) 8
- (40) The figure shows a circuit that contains three identical resistors with resistance $R = 9.0 \Omega$ each, two identical inductors with inductance $L = 2.0 mH$ each, a capacitor C , and an ideal battery with $emf \mathcal{E} = 18 V$. The current i through the battery just after the switch is closed is:
- 
- (1) $0.2 A$ (2) $4.0 A$
 (3) $0 A$ (4) $2 mA$
- (41) An electric dipole has the magnitude of its charge as q and its dipole moment is p . It is placed in a uniform electric field E . If its dipole moment is along the direction of the field, the force on it and its potential energy are respectively
 (1) $2q \cdot E$ and minimum (2) $q \cdot E$ and $p \cdot E$
 (3) Zero and minimum (4) $q \cdot E$ and maximum
- (42) The ratio of the accelerations for a solid sphere (mass m and radius R) rolling down an incline of angle θ without slipping and slipping down the incline without rolling is
 (1) $5 : 7$ (2) $2 : 3$
 (3) $2 : 5$ (4) $7 : 5$
- (43) A block of mass M slides down a rough inclined plane with constant velocity. The angle made by the inclined plane with the horizontal is θ . The magnitude of the contact force will be:
 (1) Mg (2) $Mg \cos \theta$
 (3) $\sqrt{Mg \sin \theta + Mg \cos \theta}$ (4) $Mg \sin \theta \sqrt{1 + \mu}$
- (44) When vector $\vec{A} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ is subtracted from vector $\vec{B}$, it gives a vector equal to $2\hat{j}$. Then the magnitude of vector $\vec{B}$ will be:
 (1) $\sqrt{13}$ (2) $\sqrt{33}$
 (3) $\sqrt{6}$ (4) $\sqrt{5}$
- (45) The unit of Stefan's constant σ is
 (1) $W m^{-2} K^{-1}$ (2) $W m^2 K^{-4}$
 (3) $W m^{-2} K^{-4}$ (4) $W m^{-2} K^4$

- (46) Cyclic hydrocarbon molecule A has all the carbon and hydrogen atoms in a single plane. All the carbon-carbon bonds are of the same length and are less than $1.54 \text{ \AA}$ but more than $1.34 \text{ \AA}$. The $C - C - C$ bond angle will be:
 (1) 120° (2) 180°
 (3) 100° (4) $109^\circ 28'$
- (47) The total number of σ and π bonds in 2-oxohex-4-ynoic acid is
 (1) 1 (2) 10
 (3) 18 (4) 20
- (48) Which of the following sets of quantum numbers is not possible?
 (1) $n = 3, l = 0, m = 0$ (2) $n = 3, l = 1, m = -1$
 (3) $n = 2, l = 0, m = -1$ (4) $n = 2, l = 1, m = 0$
- (49) The electrons identified by quantum numbers n and l :
 $A. n = 4, l = 1$ $B. n = 4, l = 0$
 $C. n = 3, l = 2$ $D. n = 3, l = 1$
 can be placed in order of increasing energy as :
 (1) $D < B < C < A$ (2) $D < C < B < A$
 (3) $B < D < A < C$ (4) $A < C < B < D$
- (50) Which one of the following sets of elements can be detected using sodium fusion extract?
 (1) Halogens, Nitrogen, Oxygen, Sulfur
 (2) Sulfur, Nitrogen, Phosphorous, Halogens
 (3) Nitrogen, Phosphorous, Carbon, Sulfur
 (4) Phosphorous, Oxygen, Nitrogen, Halogens
- (51) The gas phase reaction $2NO_{2(g)} \rightleftharpoons N_2O_{4(g)}$ is an exothermic reaction. The decomposition of N_2O_4 in an equilibrium mixture of $NO_{2(g)}$ and $N_2O_{4(g)}$ can be increased by:
 (1) addition of an inert gas at constant pressure
 (2) lowering the temperature
 (3) increasing the pressure
 (4) addition of an inert gas at constant volume
- (52) The value of $\log K$ for the reaction $A \rightleftharpoons B$ at 298 K is (Nearest integer).
 Given: $\Delta H^\circ = -54.07 \text{ kJ mol}^{-1}$
 $\Delta S^\circ = 10 \text{ J K}^{-1} \text{ mol}^{-1}$
 (Take $2.303 \times 8.314 \times 298 = 5705$)
 (1) 11 (2) 12
 (3) 14 (4) 10
- (53) What is the pH of the resulting solution when equal volumes of 0.1 M NaOH and 0.01 M HCl are mixed?
 (1) 2 (2) 7
 (3) 1.04 (4) 12.65

- (54) Which of the following cannot be represented by resonance structures?
 (1) Dimethyl ether (2) Nitrate anion
 (3) Carboxylate anion (4) Toluene
- (55) Which of the following may exist in enantiomorphs?
 (1) $CH_3 - CH(CH_3) - COOH$
 (2) $CH_2 = CH - CH_2 - CH_2 - CH_3$
 (3) $CH_3 - CH(NH_2) - CH_3$
 (4) $CH_3 - CH_2 - CH(NH_2) - CH_3$
- (56) Ammonium dichromate on heating gives:
 (1) Chromium oxide and ammonia
 (2) Chromic acid and nitrogen
 (3) Chromium oxide and nitrogen
 (4) Chromic acid and ammonia
- (57) The compound with the molecular formula C_3H_6O can show:
 (1) Positional isomerism
 (2) Both positional isomerism and metamerism
 (3) Metamerism
 (4) Functional group isomerism
- (58) $2.8 \times 10^{-3} \text{ mol}$ of CO_2 is left after removing 10^{21} molecules from its ' x ' mg sample. The mass of CO_2 taken initially is. Given:
 $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$. (in mg)
 (1) 196.2 (2) 98.3
 (3) 150.4 (4) 48.2

(59) Match List-I with List-II.

List-I (Elements)	List-II (Properties)
A. Cl, S	I. Elements with highest electronegativity
B. Ge, As	II. Elements with largest atomic size
C. Fr, Ra	III. Elements which show properties of f-block
D. F, O	IV. Elements with highest negative electronegativity

Choose the correct answer from the options given below:

- (1) $A - II, B - III, C - IV, D - I$
 (2) $A - III, B - II, C - I, D - IV$
 (3) $A - IV, B - III, C - II, D - I$
 (4) $A - II, B - I, C - IV, D - III$
- (60) The ground state electronic configuration of CO molecule is
 (1) $1\sigma^2 2\sigma^2 1\pi^4 3\sigma^2$
 (2) $1\sigma^2 2\sigma^2 3\sigma^2 1\pi^2 2\pi^2$
 (3) $1\sigma^2 2\sigma^2 1\pi^2 3\sigma^2 2\pi^2$
 (4) $1\sigma^2 1\pi^2 2\sigma^2 2\pi^2$

(61) Consider the equilibrium

$CO_{(g)} + 3H_{2(g)} \rightleftharpoons CH_{4(g)} + H_2O_{(g)}$. If the pressure applied over the system increases by two-fold at constant temperature, then which of the following statements are correct?

- (A) Concentration of reactants and products increases.
(B) Equilibrium will shift in the forward direction.
(C) Equilibrium constant increases since the concentration of products increases.
(D) Equilibrium constant remains unchanged as the concentration of reactants and products remain the same.

Choose the correct answer from the options given below:

- (1) (A) and (B) only
(2) (A), (B) and (D) only
(3) (B) and (C) only
(4) (A), (B) and (C) only

(62) For a spontaneous process, the incorrect statement is

- (1) $(\Delta G_{\text{system}})_{T,P} > 0$
(2) $(\Delta S_{\text{system}}) + (\Delta S_{\text{surroundings}}) > 0$
(3) $(\Delta G_{\text{system}})_{T,P} < 0$
(4) $(\Delta U_{\text{system}})_{S,V} < 0$

(63) When ψ_A and ψ_B are the wave functions of atomic orbitals, then σ^* is represented by:

- (1) $\psi_A - 2\psi_B$ (2) $\psi_A - \psi_B$
(3) $\psi_A + 2\psi_B$ (4) $\psi_A + \psi_B$

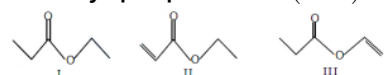
(64) The correct thermodynamic conditions for a spontaneous reaction at all temperatures are:

- (1) $\Delta H > 0$ and $\Delta S < 0$ (2) $\Delta H < 0$ and $\Delta S < 0$
(3) $\Delta H < 0$ and $\Delta S = 0$ (4) $\Delta H < 0$ and $\Delta S > 0$

(65) Which of the following conversions involves a change in both shape and hybridisation?

- (1) $H_2O \rightarrow H_3O^+$ (2) $BF_3 \rightarrow BF_4^-$
(3) $CH_4 \rightarrow C_2H_6$ (4) $NH_3 \rightarrow NH_4^+$

(66) The correct order of $C=O$ bond length in ethyl propanoate (I), ethyl propenoate (II) and ethenyl propanoate (III) is



- (1) $I > II > III$ (2) $III > II > I$
(3) $I > III > II$ (4) $II > I > III$

(67) The pair, in which ions are isoelectronic with Al^{3+} is

- (1) Br^- and Be^{2+} (2) Cl^- and Li^+
(3) S^{2-} and K^+ (4) O^{2-} and Mg^{2+}

(68) Thiol group is present in:

(1) Cysteine

(2) Methionine

(3) Cytosine

(4) Cystine

(69) If nearly $10^5 C$ liberate 1 gm equivalent of aluminium, then the amount of aluminium (equivalent weight 9) deposited through electrolysis in 20 minutes by a current of 50 A will be gm.

- (1) 0.6 (2) 0.09
(3) 5.4 (4) 10.8

(70) Given below are two statements:

Statement I: Sodium dichromate and

potassium dichromate are classified as primary standards in titrimetric analysis.

Statement II: Phenolphthalein is a weak base, therefore it dissociates in acidic medium.

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true

(71) *o*-Nitrophenol is less soluble in water than *p*- and *m*-Nitrophenols because:

(1) *o*-Nitrophenol is more volatile in steam than those of *m*- and *p*- isomers.

(2) *o*-Nitrophenol shows intramolecular H-bonding.

(3) *o*-Nitrophenol shows intermolecular H-bonding.

(4) Melting point of *o*-Nitrophenol is lower than those of *m*- and *p*- isomers.

(72) The electrolytes usually used in the

electroplating of gold and silver, respectively, are

(1) $[Au(CN)_2]^-$ and $[Ag(CN)_2]^-$

(2) $[Au(CN)_2]^-$ and $[AgCl_2]^-$

(3) $[Au(OH)_4]^-$ and $[Ag(OH)_2]^-$

(4) $[Au(NH_3)_2]^+$ and $[Ag(CN)_2]^-$

(73) For the following three esters, the order of rates of alkaline hydrolysis is:

(I) $p-NO_2-C_6H_4-COOCH_3$

(II) $p-CH_3O-C_6H_4-COOCH_3$

(III) $p-CH_3-C_6H_4-COOCH_3$

(1) $I > III > II$ (2) $II > III > I$

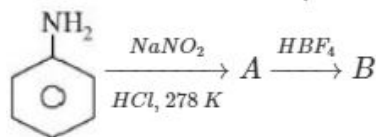
(3) $I > II > III$ (4) $III > II > I$

(74) The transition element that has the lowest enthalpy of atomisation is

- (1) Fe (2) Cu
(3) V (4) Zn

- (75) The freezing point depression constant for water is $1.86^{\circ}\text{C kg mol}^{-1}$. If 5.00 g of Na_2SO_4 is dissolved in 45.0 g of H_2O , the freezing point is lowered by 3.82°C . Calculate the van't Hoff factor for Na_2SO_4 .
- (1) 2.05 (2) 2.63
(3) 3.11 (4) 0.381
- (76) Which of the following compounds will undergo self-aldol condensation in the presence of cold dilute alkali?
- (1) $\text{C}_6\text{H}_5\text{CHO}$ (2) $\text{CH}_3\text{CH}_2\text{CHO}$
(3) CH_3CHO (4) HCHO
- (77) Predict the major product P in the following sequence of reactions—
(i) HBr , benzoyl peroxide
(ii) KCN
(iii) $\text{Na}(\text{Hg})/\text{C}_2\text{H}_5\text{OH}$
[Image: 1-methylcyclopentene reacting with the above reagents to form P]
- (1) 1-methyl-2-(aminomethyl)cyclopentane
(2) 1-methyl-1-(aminomethyl)cyclopentane
(3) 1-methyl-2-cyanocyclopentane
(4) 1-methyl-1-cyanocyclopentane
- (78) Given below are two statements :
Statement (I) : In octahedral complexes, when $\Delta_o < P$ high spin complexes are formed. When $\Delta_o > P$ low spin complexes are formed.
Statement (II) : In tetrahedral complexes because of $\Delta_t < P$, low spin complexes are rarely formed.
In the light of the above statements, choose the most appropriate answer from the options given below :
- (1) Statement I is correct but Statement II is incorrect.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are correct.
- (79) Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) : $\text{R}-\text{I}$ undergoes $\text{S}_{\text{N}}2$ reaction faster than $\text{R}-\text{Cl}$.
Reason (R) : Iodine is a better leaving group because of its large size.
In the light of the above statements, choose the correct answer from the options given below:

- (1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.
- (80) Which of the following has the least tendency to liberate H_2 from mineral acids?
- (1) Cu (2) Mn
(3) Ni (4) Zn
- (81) Acetaldehyde (CH_3CHO) reacts with:
- (1) Electrophiles only
(2) Nucleophiles only
(3) Free radicals only
(4) Both electrophiles and nucleophiles
- (82) If 250 cm^3 of an aqueous solution containing 0.73 g of a protein A is isotonic with one litre of another aqueous solution containing 1.65 g of a protein B, at 298 K , the ratio of the molecular masses of A and B is $\dots \times 10^{-2}$ (to the nearest integer).
- (1) 172 (2) 175
(3) 180 (4) 177
- (83) For a reversible reaction $\text{A} \rightleftharpoons \text{B}$, the $\Delta H_{\text{forward}} = 20\text{ kJ mol}^{-1}$. The activation energy of the uncatalysed forward reaction is 300 kJ mol^{-1} . When the reaction is catalysed keeping the reactant concentration same, the rate of the catalysed forward reaction at 27°C is found to be same as that of the uncatalysed reaction at 327°C . The activation energy of the catalysed backward reaction is $\dots\text{ kJ mol}^{-1}$.
- (1) 130 (2) 120
(3) 110 (4) 100
- (84) In the chemical reactions, the compounds A and B respectively are



- (1) nitrobenzene and fluorobenzene
(2) phenol and benzene
(3) benzene diazonium chloride and fluorobenzene
(4) nitrobenzene and chlorobenzene
- (85) Which of the following is more basic than aniline?
- (1) Benzylamine (2) Diphenylamine
(3) Triphenylamine (4) p -Nitroaniline

(86) In a particular isomer of $[Co(NH_3)_4Cl_2]^0$, the $Cl - Co - Cl$ angle is 90° . This isomer is known as:

- (1) optical isomer
- (2) *cis*-isomer
- (3) position isomer
- (4) linkage isomer

(87) Freon (dichlorodifluoromethane) is used:

- (1) As local anaesthetic
- (2) For dissolving impurities in metallurgical process
- (3) In refrigerators
- (4) In printing industry

(88) Phenol is heated with a solution of a mixture of KBr and $KBrO_3$. The major product obtained in the above reaction is:

- (1) 2- Bromophenol
- (2) 3- Bromophenol
- (3) 4- Bromophenol
- (4) 2, 4, 6- Tribromophenol

(89) Time required for 99.9% completion of a first order reaction is times the time required for completion of 90% reaction. (nearest integer).

- (1) 5
- (2) 4
- (3) 3
- (4) 8

(90) $2 NO_{(g)} + Cl_{2(g)} \rightleftharpoons 2 NOCl_{(g)}$

This reaction was studied at $-10^\circ C$ and the following data was obtained:

Run	$[NO]_0$	$[Cl_2]_0$	r_0
1	0.10	0.10	0.18
2	0.10	0.20	0.35
3	0.20	0.20	1.40

$[NO]_0$ and $[Cl_2]_0$ are the initial concentrations and r_0 is the initial reaction rate.

The overall order of the reaction is (Round off to the Nearest Integer).

- (1) 1
- (2) 2
- (3) 3
- (4) 4

Biology - Section A (MCQ)

(91) Biradial symmetry and lack of cnidoblasts are the characteristics of

- (1) Hydra and starfish
- (2) starfish and sea anemone
- (3) Ctenoplane and Beroe
- (4) Aurelia and Paramecium

(92) A system of classification in which a large number of traits are considered is

- (1) Artificial system
- (2) Synthetic system
- (3) Natural system
- (4) Phylogenetic system

(93) Which of the following is not a true amphibian?

- (1) Salamander
- (2) Toad
- (3) Turtle
- (4) Frog

(94) A vein differs from an artery in having

- (1) Narrow lumen
- (2) Strong cuticular and muscular wall
- (3) Valves to control direction of flow
- (4) Dark pigmented wall

(95) Which one of the following does not act as a neurotransmitter?

- (1) Cortisone
- (2) Acetylcholine
- (3) Epinephrine
- (4) Norepinephrine

(96) In gymnosperms, the pollen chamber represents

- (1) a cavity in the ovule in which pollen grains are stored after pollination
- (2) an opening in the megagametophyte through which the pollen tube approaches the egg
- (3) the microsporangium in which pollen grains develop
- (4) a cell in the pollen grain in which the sperms are formed

(97) Which one is the most abundant protein in the animal world?

- (1) Trypsin
- (2) Haemoglobin
- (3) Collagen
- (4) Insulin

(98) A typical fat molecule is made up of

- (1) one glycerol and one fatty acid molecule
- (2) three glycerol and three fatty acid molecules
- (3) three glycerol molecules and one fatty acid molecule
- (4) one glycerol and three fatty acid molecules

(99) In tissue culture experiments, leaf mesophyll cells are put in a culture medium to form callus. This phenomenon may be called as

- (1) Senescence
- (2) Differentiation
- (3) Dedifferentiation
- (4) Development

(100) The basic functional unit of the human kidney is:

- (1) Nephron
- (2) Pyramid
- (3) Nephridia
- (4) Henle's loop

- (101) Compound apocarpous gynoecium is found in
 (1) Lily (2) Hollyhock
 (3) Lotus/Ranunculus (4) Pumpkin
- (102) All fungi can be called as
 (1) Heterotrophs
 (2) Autotrophs
 (3) Saprophytes
 (4) Parasites
- (103) Synapsis occurs between
 (1) *mRNA* and ribosomes
 (2) spindle fibres and centromere
 (3) two homologous chromosomes
 (4) a male and a female gamete.
- (104) Apical dominance in plants is caused by
 (1) High concentration of auxins in the terminal bud
 (2) High concentration of gibberellins in the apical bud
 (3) High concentration of auxins in the lateral bud
 (4) Absence of auxins and gibberellins in the apical bud
- (105) Which of the following is a vascular, non-flowering plant?
 (1) Equisetum (2) Ginkgo
 (3) Marchantia (4) Cedrus
- (106) The complex formed by a pair of synapsed homologous chromosomes is called
 (1) bivalent
 (2) axoneme
 (3) equatorial plate
 (4) kinetochore
- (107) Select the incorrect statements with respect to Cyclostomes:
 (a) They lack scales and paired fins.
 (b) They have a circular mouth with jaws.
 (c) They bear 6 – 15 pairs of gills.
 (d) They migrate to deep sea for spawning.
 Choose the most appropriate answer from the options given below:
 (1) (a) and (b) only (2) (b) and (c) only
 (3) (b) and (d) only (4) (a) and (d) only
- (108) Which of the following cellular organelles are membrane-bound?
 (1) Endoplasmic reticulum, ribosomes, and nuclei
 (2) Lysosomes, Golgi apparatus, and mitochondria
 (3) Nuclei, ribosomes, and mitochondria
 (4) Chromosomes, ribosomes, and endoplasmic reticulum
- (109) A protoplast is a cell that is:
 (1) Without a plasma membrane
 (2) Without a nucleus
 (3) Capable of division
 (4) Without a cell wall
- (110) The maximum amount of electrolytes and water (70 – 80 %) from the glomerular filtrate is reabsorbed in which part of the nephron?
 (1) Ascending limb of loop of Henle
 (2) Distal convoluted tubule
 (3) Proximal convoluted tubule
 (4) Descending limb of loop of Henle
- (111) Which of the following statements is incorrect?
 (1) During aerobic respiration, the role of oxygen is limited to the terminal stage.
 (2) In *ETC* (Electron Transport Chain), one molecule of $NADH + H^+$ gives rise to 2 *ATP* molecules, and one $FADH_2$ gives rise to 3 *ATP* molecules.
 (3) *ATP* is synthesized through complex V.
 (4) Oxidation-reduction reactions produce a proton gradient in respiration.
- (112) Black (stem) rust of wheat is caused by
 (1) *Alternaria solani* (2) *Ustilago nuda*
 (3) *Puccinia graminis* (4) *Xanthomonas oryzae*
- (113) The pacemaker of the heart is known as
 (1) *AV* node (2) Bundle of His
 (3) *SA* node (4) Purkinje fibers
- (114) Which of the following statements is incorrect regarding *Cycas*?
 (1) Its xylem has vessels.
 (2) It has circinate vernation.
 (3) It does not have a well-organized female flower.
 (4) Its roots possess some blue-green algae.
- (115) Palisade parenchyma is absent in leaves of
 (1) mustard (2) soybean
 (3) gram (4) Sorghum

- (116) One scientist cultured *Cladophora* in a suspension of *Azotobacter* and illuminated the culture by splitting light through a prism. He observed that bacteria accumulated mainly in the region of
- (1) Violet and green light
 - (2) Indigo and green light
 - (3) Orange and yellow light
 - (4) Blue and red light
- (117) The technical term used for the androecium in a flower of China rose (*Hibiscus rosa sinensis*) is
- (1) monadelphous
 - (2) diadelphous
 - (3) polyandrous
 - (4) polyadelphous
- (118) The intrafascicular cambium and interfascicular cambium are
- (1) parts of secondary phloem and xylem.
 - (2) parts of the pericycle.
 - (3) lateral meristems.
 - (4) apical meristems.
- (119) Which one of the following correctly describes the location of some body parts in the earthworm *Pheretima*?
- (1) Four pairs of spermathecae in 4th – 7th segments
 - (2) One pair of ovaries attached at intersegmental septum of 14th and 15th segments
 - (3) Two pairs of testes in 10th and 11th segments
 - (4) Two pairs of accessory glands in 16th – 18th segments
- (120) Match List-I with List-II.
- | List-I | List-II |
|----------------------------|--------------------------------|
| A. Heart | I. Erythropoietin |
| B. Kidney | II. Aldosterone |
| C. Gastro-intestinal tract | III. Atrial natriuretic factor |
| D. Adrenal Cortex | IV. Secretin |
- Choose the correct answer from the options given below:
- (1) A – II, B – I, C – III, D – IV
 - (2) A – IV, B – III, C – II, D – I
 - (3) A – I, B – III, C – IV, D – II
 - (4) A – III, B – I, C – IV, D – II
- (121) The enzyme required for the conduction of

nerve impulses across the synapse is:

- (1) Peroxidase
 - (2) Choline acetylase
 - (3) Ascorbic acid oxidase
 - (4) Succinic dehydrogenase
- (122) Mitosis is the process by which eukaryotic cells
- (1) Expose the genes for protein synthesis
 - (2) Become specialized in structure and function
 - (3) Multiply
 - (4) Grow
- (123) In the eastern states of India, rice production decreases during the monsoon season. Which factor is primarily responsible for this?
- (1) CO₂
 - (2) Light
 - (3) Temperature
 - (4) Water
- (124) Ligament is a
- (1) Modified white fibrous tissue
 - (2) Modified yellow elastic fibrous tissue
 - (3) Inelastic white fibrous tissue
 - (4) None of these
- (125) In germinating castor seeds, the R.Q. is
- (1) One
 - (2) More than one
 - (3) Less than one
 - (4) Zero
- (126) In which stage of meiosis does the splitting of the centromere occur?
- (1) Metaphase-I
 - (2) Metaphase-II
 - (3) Anaphase-II
 - (4) Telophase-II
- (127) In paddy fields, cyanobacteria are found in the vegetative parts of
- (1) Pinus
 - (2) Cycas
 - (3) Equisetum
 - (4) Psilotum
- (128) Tracheae of cockroach and mammal are similar in having
- (1) Paired nature
 - (2) Non-collapsible walls
 - (3) Ciliated inner lining
 - (4) Origin from head
- (129) The pivot joint between atlas and axis is a type of
- (1) cartilaginous joint
 - (2) synovial joint
 - (3) saddle joint
 - (4) fibrous joint.

- (130) Which of the following characteristics is incorrect with respect to the cockroach?
- (1) A ring of gastric caeca is present at the junction of midgut and hindgut.
 - (2) Hypopharynx lies within the cavity enclosed by the mouthparts.
 - (3) In females, 7th – 9th sterna together form a genital pouch.
 - (4) 10th abdominal segment in both sexes bears a pair of anal cerci.
- (131) Bulk of carbon dioxide (CO_2) released from body tissues into the blood is present as
- (1) 70% carbamino-haemoglobin and 30% as bicarbonate
 - (2) carbamino-haemoglobin in *RBCs*
 - (3) bicarbonate in blood plasma and *RBCs*
 - (4) free CO_2 in blood plasma
- (132) Two plants can be conclusively said to belong to the same species if they
- (1) have more than 90 percent similar genes
 - (2) look similar and possess identical secondary metabolites
 - (3) have same number of chromosomes
 - (4) can reproduce freely with each other and form seeds.
- (133) Name the ion responsible for unmasking of active sites for myosin for cross-bridge activity during muscle contraction.
- (1) Calcium
 - (2) Magnesium
 - (3) Sodium
 - (4) Potassium
- (134) People who have migrated from the plains to an area adjoining Rohtang Pass about six months back:
- (1) have more *RBCs* and their haemoglobin has a lower binding affinity to O_2
 - (2) are not physically fit to play games like football
 - (3) suffer from altitude sickness with symptoms like nausea, fatigue, etc.
 - (4) have the usual *RBC* count but their haemoglobin has very high binding affinity to O_2
- (135) Which era is known as the 'Age of Reptiles'?
- (1) Cenozoic Era
 - (2) Mesozoic Era
 - (3) Paleozoic Era
 - (4) Archeozoic Era
- (136) Sporeine was developed in
- (1) Germany
 - (2) France
 - (3) U.S.A.
 - (4) Russia
- (137) Which of the following ecosystems has the highest primary productivity?
- (1) Grasslands
 - (2) Coral reefs
 - (3) Mangroves
 - (4) Tropical rainforests
- (138) Tubectomy is a method of sterilization in which
- (1) small part of the Fallopian tube is removed or tied up
 - (2) ovaries are removed surgically
 - (3) small part of vas deferens is removed or tied up
 - (4) uterus is removed surgically.
- (139) The one aspect which is not a salient feature of the genetic code is its being
- (1) degenerate
 - (2) ambiguous
 - (3) universal
 - (4) specific
- (140) Which one of the following statements is correct?
- (1) Endothecium produces the microspores.
 - (2) Tapetum nourishes the developing pollen.
 - (3) Hard outer layer of pollen is called intine.
 - (4) Sporogenous tissue is haploid.
- (141) Which ecosystem has the maximum biomass?
- (1) Grassland ecosystem
 - (2) Pond ecosystem
 - (3) Lake ecosystem
 - (4) Forest ecosystem
- (142) The sperm penetrates the ovum mainly by:
- (1) Mechanically
 - (2) Chemically
 - (3) Electrostatically
 - (4) Thermally
- (143) Embryological support for evolution was proposed by
- (1) Alfred Wallace
 - (2) Ernst Haeckel
 - (3) Karl Ernst von Baer
 - (4) Charles Darwin
- (144) Animals have the innate ability to escape from predation. Examples for the same are given below. Select the incorrect example.

- (1) Colour change in chameleon
- (2) Enlargement of body size by swallowing air in puffer fish
- (3) Poison fangs in snakes
- (4) Melanism in moths

(145) Which fungal disease is transmitted through seeds and flowers?

- (1) Loose smut of wheat
- (2) Maize stunt
- (3) Covered smut of barley
- (4) Soft rot of potato

(146) Foetal sex can be determined by examining cells from the amniotic fluid by looking for

- (1) Barr bodies
- (2) Autosomes
- (3) Chiasmata
- (4) Kinetochore

(147) The condition of sex chromosomes in a male child with Down's syndrome will be:

- (1) XY
- (2) XXY
- (3) XX
- (4) XO

(148) Which of the following is an in situ conservation method?

- (1) (1) Seed Banks
- (2) (2) Sacred Groves
- (3) (3) Botanical Gardens
- (4) (4) Wildlife Safari Parks

(149) *E.coli* cells with a mutated Z gene of the *lac* operon cannot grow in a medium containing only lactose as the source of energy because:

- (1) In the presence of glucose, *E.coli* cells do not utilize lactose.
- (2) They cannot transport lactose from the medium into the cell.
- (3) The *lac* operon is constitutively active in these cells.
- (4) They cannot synthesize functional β -galactosidase.

(150) The primitive prokaryotes responsible for the production of biogas from the dung of ruminant animals include the:

- (1) methanogens
- (2) eubacteria
- (3) halophiles
- (4) thermoacidophiles

(151) Match List-I with List-II : Choose the correct answer from the options given below :

List-I	List-II
A. Down's syndrome	I. 11 th chromosome
B. α -Thalassemia	II. X chromosome
C. β -Thalassemia	III. 21 st chromosome
D. Klinefelter's syndrome	IV. 16 th chromosome

(1) $A - II, B - III, C - IV, D - I$

(2) $A - III, B - IV, C - I, D - II$

(3) $A - IV, B - I, C - II, D - III$

(4) $A - I, B - II, C - III, D - IV$

(152) If a bamboo plant is growing in a forest, what will be its trophic level?

- (1) First
- (2) Second
- (3) Third
- (4) Fourth

(153) Upon exposure to *UV* radiation, *DNA* stained with ethidium bromide will show:

- (1) Bright orange colour
- (2) Bright red colour
- (3) Bright blue colour
- (4) Bright yellow colour

(154) Humans have an acquired immune system that produces antibodies to neutralize pathogens. Still, the innate immune system is present at the time of birth because it

- (1) is very specific and uses different macrophages.
- (2) produces memory cells for mounting a fast secondary response.
- (3) has natural killer cells which can phagocytose and destroy microbes.
- (4) provides passive immunity.

(155) Which part of the tobacco plant is infected by *Meloidogyne incognita*?

- (1) Stem
- (2) Root
- (3) Flower
- (4) Leaf

(156) The shortest gestation period is seen in

- (1) Man
- (2) Elephant
- (3) Cat
- (4) Mouse

(157) *PCR* and Restriction Fragment Length Polymorphism (*RFLP*) are techniques used for.....

- (1) Study of enzymes
- (2) Genetic mutation
- (3) *DNA* sequencing
- (4) *DNA* fingerprinting

(158) Which extraembryonic membrane in humans prevents desiccation of the embryo inside the uterus?

- (1) Yolk sac
- (2) Amnion
- (3) Chorion
- (4) Allantois

(159) The age pyramid with a broad base indicates:

- (1) high percentage of old individuals
- (2) low percentage of young individuals
- (3) a stable population
- (4) high percentage of young individuals

(160) Alleles are

- (1) different molecular forms of a gene
- (2) heterozygotes
- (3) different phenotype
- (4) true breeding homozygotes

(161) Severe Acute Respiratory Syndrome (SARS):

- (1) Is caused by a variant of Pneumococcus
- (2) Is caused by a variant of the common cold virus (coronavirus)
- (3) Is an acute form of asthma
- (4) Affects non-vegetarians much faster than the vegetarians

(162) Mycorrhizae are an example of

- (1) amensalism
- (2) antibiosis
- (3) mutualism
- (4) fungistasis

(163) The human protein named

α - 1-antitrypsin, obtained from transgenic animals, is used for the treatment of

- (1) Alzheimer's disease (2) Rheumatoid arthritis
- (3) Emphysema (4) Cystic fibrosis

(164) The hilum is a scar on the

- (1) fruit, where style was present
- (2) seed, where micropyle was present
- (3) seed, where funicle was attached
- (4) fruit, where it was attached to pedicel.

(165) Which of the following subdivisions of biology deals with the study of fossils?

- (1) Ecology
- (2) Ornithology
- (3) Ethology
- (4) Palaeontology

(166) The presence of gill slits in the embryos of all vertebrates supports which theory?

(1) Biogenesis

(2) Recapitulation theory

(3) Metamorphosis

(4) Organic evolution

(167) The microbe used for the biocontrol of pest butterfly caterpillars is:

- (1) *Saccharomyces cerevisiae*
- (2) *Bacillus thuringiensis*
- (3) *Streptococcus sp.*
- (4) *Trichoderma sp.*

(168) *Escherichia coli* (*E.coli*) is widely used in biological research because:

- (1) It can be easily cultured.
- (2) It is easily available.
- (3) Its maintenance is simple.
- (4) It can easily multiply in the host.

(169) Match the following enzymes with their functions:

(a) Restriction endonuclease	(i) Joins the DNA fragments
(b) Restriction exonuclease	(ii) Extends primers on genomic DNA template
(c) DNA ligase	(iii) Cuts DNA at specific positions
(d) Taq polymerase	(iv) Removes nucleotides from the ends of DNA

Select the correct option from the following:

- (1) $a - iii, b - i, c - iv, d - ii$
- (2) $a - iii, b - iv, c - i, d - ii$
- (3) $a - iv, b - iii, c - i, d - ii$
- (4) $a - ii, b - iv, c - i, d - iii$

(170) What is the figure given below showing in particular?



- (1) Ovarian cancer
- (2) Uterine cancer
- (3) Tubectomy
- (4) Vasectomy

(171) Evolution of different species in a given area starting from a point and spreading to other geographical areas is known as

- (1) adaptive radiation
- (2) natural selection
- (3) migration
- (4) divergent evolution

(172) If the forest area is reduced by half, what is the long-term consequence likely to be?

- (1) Tribals living in this area will die of starvation.
- (2) Cattle in this and surrounding areas will die due to lack of fodder.
- (3) A large area will turn into a desert.
- (4) Crop breeding programs will be adversely affected due to the loss of germplasm diversity.

(173) Both, autogamy and geitonogamy are prevented in

- (1) papaya (2) cucumber
- (3) castor (4) maize

(174) Haploid plants are obtained from which of the following?

- (1) Leaves (2) Root tips
- (3) Pollen grains (4) Buds

(175) The genes, which are confined to the differential region of the *Y* chromosome only, are called:

- (1) Mutant
- (2) Autosomal
- (3) Holandric
- (4) Completely sex-linked

(176) Which of the following restriction enzymes produces blunt ends?

- (1) *Xho* – I (2) *Hind* – III
- (3) *Sal* – I (4) *Eco* – RV

(177) Match List I with List II.

List I	List II
A. Non-medicated IUD	I. Multiload 375
B. Copper releasing IUD	II. Progestogens
C. Hormone releasing IUD	III. Lippes loop
D. Implants	IV. LNG – 20

Choose the correct answer from the options given below:

- (1) A – I, B – III, C – IV, D – II
- (2) A – IV, B – I, C – II, D – III
- (3) A – III, B – I, C – IV, D – II
- (4) A – III, B – I, C – II, D – IV

(178) Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): When a particular restriction enzyme cuts a strand of *DNA*, overhanging stretches or sticky ends are formed.

Reason (R): Some restriction enzymes cut the strand of *DNA* a little away from the centre of the palindromic site.

In the light of the above statements, choose the correct answer from the options given below:

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
- (3) (A) is correct but (R) is not correct
- (4) (A) is not correct but (R) is correct

(179) Match the items in Column-I with those in Column-II:

Column-I	Column-II
(a) Herbivores-Plants	(i) Commensalism
(b) Mycorrhiza-Plants	(ii) Mutualism
(c) Sheep-Cattle	(iii) Predation
(d) Orchid-Tree	(iv) Competition

Select the correct option from the following:

- (1) (a) – (i), (b) – (iii), (c) – (iv), (d) – (ii)
- (2) (a) – (iv), (b) – (ii), (c) – (i), (d) – (iii)
- (3) (a) – (iii), (b) – (ii), (c) – (iv), (d) – (i)
- (4) (a) – (ii), (b) – (i), (c) – (iii), (d) – (iv)

(180) Which one of the following is categorised as a parasite in the true sense?

- (1) The female *Anopheles* bites and sucks blood from humans.
- (2) Human foetus developing inside the uterus draws nourishment from the mother.
- (3) Head louse living on the human scalp as well as laying eggs on human hair.
- (4) The cuckoo (koel) lays its eggs in a crow's nest.